

Mathematics for Biomedical Engineering

L1:

Introduction

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- To understand basic concepts of algebra, logarithms and exponentials, and trigonometry

1 Basic Algebra

2 Logarithms and Exponentials

3 Trigonometry

1 Basic Algebra

2 Logarithms and Exponentials

3 Trigonometry

Basic Algebra

To be able to apply mathematics to solve engineering problems it is necessary to introduce symbols to represent physical quantities.

Algebra is the study of mathematical symbols and the rules for manipulating these symbols. In algebra, numbers are often represented by symbols called **variables**. We use **functions** to describe how a variable changes as a consequence of another changing.

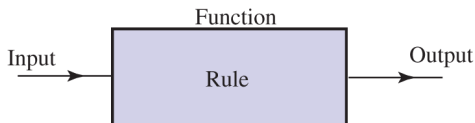
A firm understanding of the following algebraic concepts and techniques is essential for this module:

- Factorisation (quadratic expressions)
- Algebraic fractions (addition, subtraction, multiplication, division)
- Functions (odd, even, periodic)
- Equation of a line
- Polynomials
- Polynomial equations
- Solving quadratic equations
- Solving simultaneous equations
- Inequalities

You can revise and practise these concepts and techniques by reading Chapters 5 to 7 of the main textbook. You can also have a look at the *Maths First-Aid Kit* (download it from [here](#))

Functions

A **function** is a rule that operates on an input variable, x , and produces a single output variable, y , from that input.



Common notation: $y = f(x)$.

The input variable is also called the **independent variable** and the output variable is also called the **dependent variable**

The set of x values used as input to the function is called the **domain** of the function. The set of values that y takes as x is varied is called the **range** of the function.

Example: What is the domain and range of the functions $y = x^2$ and $y = \frac{1}{x-7}$?

Partial Fractions

A complicated algebraic fraction can be broken down into the sum of simpler fractions, which we call **partial fractions**.

Methodology:

- 1 Factorise the denominator
- 2 Write as many partial fractions as factors; each multiplied by an unknown constant
- 3 Find the value of the unknown constants

Example: Express $\frac{7x+10}{2x^2+5x+3}$ as its partial fractions.

Partial Fractions – Important Considerations

- A linear factor $ax + b$ produces a partial fraction of the form

$$\frac{A}{ax + b}$$

- Repeated linear factors $ax + b$ produce partial fractions of the form

$$\frac{A}{ax + b} + \frac{B}{(ax + b)^2}$$

- A quadratic factor of the form $ax^2 + bx + c$ produces a partial fraction of the form

$$\frac{Ax + B}{ax^2 + bx + c}$$

- If a fraction is **improper** (the degree of the numerator, n , is greater than the degree of the denominator, d), then an additional term must be included taking the form of a polynomial of degree $n - d$

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The Exponential Function

It has widespread application in many areas of science and engineering; e.g. in expansion of materials, laws of cooling, radioactive decay, and the discharge of a capacitor.

An exponential function has the form

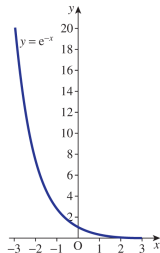
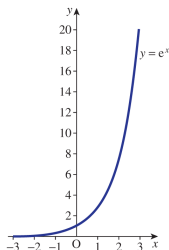
$$y = a^x$$

where a is a positive constant called the **base**.

The most commonly used exponential function, commonly called **the exponential function**, is

$$y = e^x$$

where the base e is the exponential constant whose value is $e = 2.71828182 \dots$



Hyperbolic Functions

They are defined using the exponential function e^x .

Key definitions

$$\text{hyperbolic sine: } \sinh x = \frac{e^x - e^{-x}}{2}$$

$$\text{hyperbolic cosine: } \cosh x = \frac{e^x + e^{-x}}{2}$$

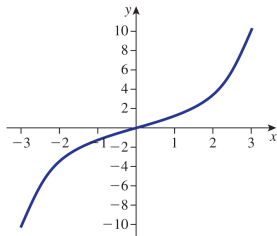
$$\text{hyperbolic tangent: } \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\text{cosech } x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

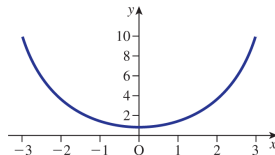
$$\text{sech } x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

$$\text{coth } x = \frac{1}{\tanh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

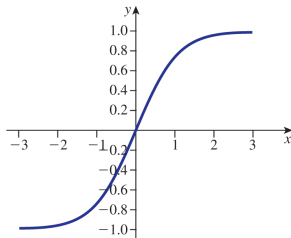
Hyperbolic Functions – Plots



(a)



(b)



(c)

Figure: (a) $y = \sinh x$, (b) $y = \cosh x$, (c) $y = \tanh x$.

Common Hyperbolic Identities

$$\begin{aligned}e^x &= \cosh x + \sinh x \\e^{-x} &= \cosh x - \sinh x \\\cosh^2 x - \sinh^2 x &= 1 \\1 - \tanh^2 x &= \operatorname{sech}^2 x \\\coth^2 x - 1 &= \operatorname{cosech}^2 x \\\sinh(x \pm y) &= \sinh x \cosh y \pm \cosh x \sinh y \\\cosh(x \pm y) &= \cosh x \cosh y \pm \sinh x \sinh y \\\sinh 2x &= 2 \sinh x \cosh x \\\cosh 2x &= \cosh^2 x + \sinh^2 x \\\cosh^2 x &= \frac{\cosh 2x + 1}{2} \\\sinh^2 x &= \frac{\cosh 2x - 1}{2}\end{aligned}$$

Note that $\cosh^2 x$ means $(\cosh x)^2$ and $\sinh^2 x$ means $(\sinh x)^2$.

Logarithms

Logarithms are an alternative way of writing expressions that involve powers. They are the inverse function to exponentiation.

If a is a positive constant and

$$N = a^x$$

then

$$\log_a N = x$$

The number a is called the **base** of the logarithm. Usually $a = 10$ or $a = e$.

Logarithms to the base 10 or e

$$\text{If } N = 10^x \text{ then } x = \log_{10} N = \log N$$

$$\text{If } N = e^x \text{ then } x = \log_e N = \ln N$$

Example: Calculate the following logarithms: $\log 1000$, $\log 0.01$, $\ln e^x$

Logarithms to any base

$$\log_a N = \frac{\log_b N}{\log_b a}$$

Example: Evaluate $\log_3 40$

Laws of Logarithms

They hold true for any base.

Important Identities

$$\log AB = \log A + \log B$$

$$\log A/B = \log A - \log B$$

$$\log A^n = n \log A$$

Note that from the second law we have

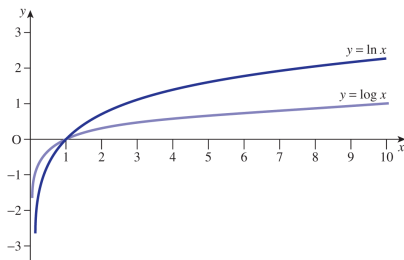
$$\log 1 = 0$$

Example: Express $\ln t^5 - 3 \ln t^2$ as a single logarithm.

Logarithmic Functions

They are defined as

$$y = \log x \text{ and } y = \ln x \text{ for } x > 0$$



The domain of both functions is $x > 0$ and the range is $(-\infty, \infty)$.

1 Basic Algebra

2 Logarithms and Exponentials

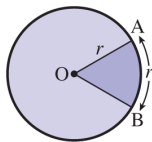
3 Trigonometry

Angles

Angles measure the amount through which a line or object has been turned. They can be measured in **degrees** or **radians**.

Definition of Radian

1 radian = the angle subtended at centre O by an arc AB whose length is one radius, r

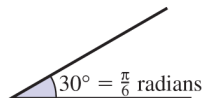


An arc length of $2\pi r$ is the entire circumference. The circumference subtends a complete revolution at the centre, that is 360° , and so

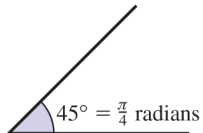
$$2\pi \text{ radians} = 360^\circ, \quad \pi \text{ radians} = 180^\circ$$

Some Common Angles

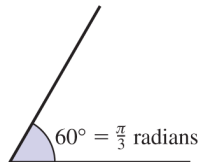
Commonly angles measured in radians are expressed as multiples of π .



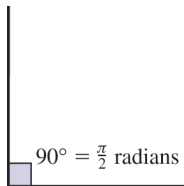
(a)



(b)



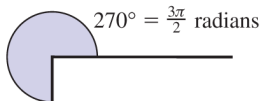
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(d)



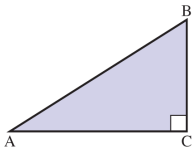
(e)



(f)

The Trigonometrical Ratios

They are defined based on a right-angled triangle.



Key definitions

$$\sin A = \frac{\text{length of side opposite}}{\text{length of hypotenuse}} = \frac{BC}{AB}$$

$$\cos A = \frac{\text{length of side adjacent}}{\text{length of hypotenuse}} = \frac{AC}{AB}$$

$$\tan A = \frac{\text{length of side opposite}}{\text{length of side adjacent}} = \frac{BC}{AC}$$

Using the same definitions, for the angle at B we have:

$$\sin B = \frac{AC}{AB}, \quad \cos B = \frac{BC}{AB}, \quad \tan B = \frac{AC}{BC}.$$

Some Properties of the Trigonometrical Ratios

$$\tan A = \frac{\sin A}{\cos A}, \quad \tan B = \frac{\sin B}{\cos B},$$

$$\sin A = \cos B = \cos (90^\circ - A), \quad \cos A = \sin B = \sin (90^\circ - A),$$

$$\tan A = \frac{1}{\tan B} = \frac{1}{\tan (90^\circ - A)}.$$

Secant, Cosecant and Cotangent Ratios

These ratios are the reciprocals of the cosine, sine and tangent ratios.

Key definitions

$$\sec A = \frac{1}{\cos A}$$

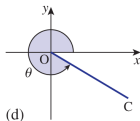
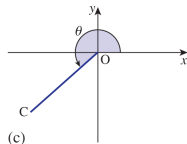
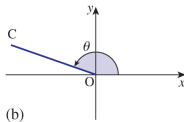
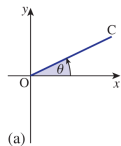
$$\operatorname{cosec} A = \frac{1}{\sin A}$$

$$\cot A = \frac{1}{\tan A}$$

The Trigonometrical Ratios in All Quadrants

To define the trigonometrical ratios of angles greater than 90° we need to use the x - y plane.

We define the trigonometrical ratios in terms of the x and y projections of a rotating arm, OC .



Key definitions

$$\sin \theta = \frac{y \text{ projection of } OC}{OC}$$

$$\cos \theta = \frac{x \text{ projection of } OC}{OC}$$

$$\tan \theta = \frac{y \text{ projection of } OC}{x \text{ projection of } OC}$$

Figure: The arm OC rotates anticlockwise into each of the four quadrants.

Note that $\sin \theta = \sin (\theta \pm n 360^\circ)$ and $\cos \theta = \cos (\theta \pm n 360^\circ)$, $n = 1, 2, 3, \dots$
 and $\tan \theta = \tan (\theta \pm n 180^\circ)$, $n = 1, 2, 3, \dots$

Trigonometrical Functions

They are all **periodic functions**; i.e. their y values are repeated at regular intervals.

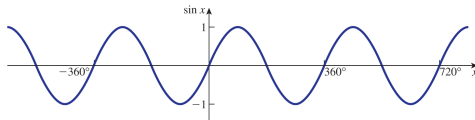


Figure: The function $y = \sin x$ has a period of 360° or 2π .

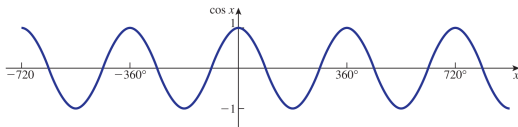


Figure: The function $y = \cos x$ has a period of 360° or 2π .

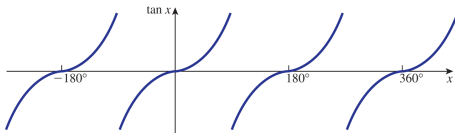


Figure: The function $y = \tan x$ has a period of 180° or π .

Common Trigonometrical Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\frac{\sin A}{\cos A} = \tan A$$

$$\sin(A + B) = \sin A \cos B + \sin B \cos A$$

$$\sin(A - B) = \sin A \cos B - \sin B \cos A$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1 = \cos^2 A - \sin^2 A$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

$$\sin^2 A = \frac{1}{2}(1 - \cos 2A)$$

$$\cos^2 A = \frac{1}{2}(1 + \cos 2A)$$

Note that $\cos^2 A$ means $(\cos A)^2$ and $\sin^2 A$ means $(\sin A)^2$.

Engineering Waves

Consider the waves $y = A \sin \omega t$ and $y = A \cos \omega t$, with t the time measured in seconds.

We call A the **amplitude** of the wave and ω the **angular frequency** of the wave.

Note that ω has units of radians per second.

The period of a wave

The **period**, T , of the wave is the time to complete one full cycle.

Given that the sine and cosine functions complete one full cycle as the angle increases by 2π radians, for both $y = A \sin \omega t$ and $y = A \cos \omega t$ we have

$$T = \frac{2\pi}{\omega}$$

The frequency of a wave

The **frequency**, f , of the wave is the number of cycles completed in 1 second. It is measured in hertz (Hz).

For both $y = A \sin \omega t$ and $y = A \cos \omega t$, 1 cycle is completed in $\frac{2\pi}{\omega}$ seconds. Hence, ω cycles are completed in 2π seconds and $\frac{\omega}{2\pi}$ cycles in 1 second. Thus,

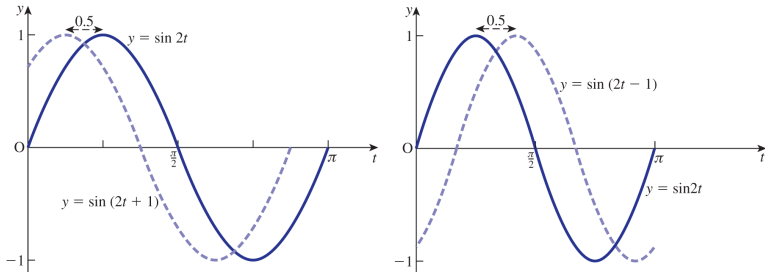
$$f = \frac{\omega}{2\pi} = \frac{1}{T}$$

Phase and Time Displacement of a Wave

Consider the waves $y = A \sin(\omega t + \alpha)$ and $y = A \cos(\omega t + \alpha)$, with t the time measured in seconds.

The **phase angle**, α , has the effect of moving the wave to either the left or the right.

Examples:



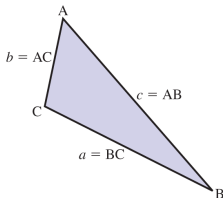
We say that $\sin(2t + 1)$ **leads** $\sin 2t$ by 0.5 s and $\sin(2t - 1)$ **lags** $\sin 2t$ by 0.5 s.

Time displacement of a wave

The **time displacement** of the waves $y = A \sin(\omega t + \alpha)$ and $y = A \cos(\omega t + \alpha)$ is $\frac{\alpha}{\omega}$ seconds.

The Sine and Cosine Rules

Consider the generic triangle ABC as shown in the figure.



Sine Rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Example: Solve the triangle ABC given $A = 46^\circ$, $B = 32^\circ$ and $BC = 13$ cm.

Cosine Rule

$$a^2 = b^2 + c^2 - 2bc \cos A \quad b^2 = a^2 + c^2 - 2ac \cos B \quad c^2 = a^2 + b^2 - 2ab \cos C$$

Example: Solve the triangle ABC given $B = 71^\circ$, $AB = 17.3$ cm and $BC = 23.9$ cm.

The Sine and Cosine Rules – When to Use Them?

They can both be applied to any triangle, including those that are not right angled.

The **sine rule** can be used to solve a triangle when we are given either:

- One side and two angles
- Two sides and one angle that is not included by the given sides.

The **cosine rule** can be used to solve a triangle when we are given either:

- Three sides
- Two sides and the included angle

What We Learnt in Topic 1

- We have reviewed basic concepts of algebra, logarithms and exponentials, and trigonometry